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Bahir Dar University

Crop Production and Management



Tuesday, September 3, 2024

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1



Introduction to Crop Production



- **Production:** - the act or an instance of producing; the activity of being producing a total yield
- **Crops:** - refers to plants sown and harvested by man for economic purpose (food, feed, fiber, sugar and for other purposes).
- **Crop production** is the practice/art/ of growing and harvesting of crops.
- It is a complex operation which consists of
 - site selection and clearing of vegetation,
 - seed bed preparation,
 - sowing seeds,
 - nutritional management (fertilizer and water),
 - controlling insect pest, diseases and weeds and
 - harvesting, processing and storing seeds.
- **crop production** is basically the conversation of the environmental inputs of solar energy, carbon dioxide, water and soil nutrients in to economic end products of human and animal foods staffs, fodder, fiber, wood and other industrial production.



Introduction to Crop Production



❖ Aims of crop production

- Maintain food self sufficiency
- Support the local populations in productive work
- Be source of raw materials for the development of industry
- Produce export crops in order to earn foreign currency
- Provide finance for the development of industry (through taxation) and etc



Introduction to Crop Production



❖ Importance of crop production

- Humans cannot survive without crop plants. It is used as:
 - ❖ Food for human and feed for livestock.
 - ❖ Wood and wood products from fiber crops: construction, furniture, fuel and paper.
 - ❖ Textiles from fiber producing plants like cotton, sisal, flax
 - ❖ Drugs and medicines examples tobacco, coffee
 - ❖ Prevent erosion
 - ❖ Perfumes and spices e.g. perfume-rose, spice- pepper and ginger,
 - ❖ Aesthetic purpose (for great pleasure, relaxation) and source of oxygen



FOOD

- It is the basic requirement of life to survive.
- It is made up of various kind of energy rich substances like carbohydrates, proteins, fats, vitamins and minerals, which are known as nutrients.



CONT'D



FUNCTIONS OF FOOD

- Food provide energy for various metabolic activities.
- It helps in growth and development of body.
- It helps in the replacement of worn out tissues, repair of damaged cells and healing of wounds.
- It provide protection against diseases.

Sources of foods:

- Cereals
- Pulses
- Oilseed crops
- Fruits & Vegetables
- Roots & tuber crops



World Population and Food Situation



World great famines in chronological order

No.	Years (AD)	Famine Area/Country	Magnitude of the Famine
1	879	Universal world famine	Most part of the world
2	1125	Germany	Half of the total population diminished
3	1505	Hungary	Devastated the country seriously
4	1700-1850	European countries	Common in most European countries
5	1870-1871	Persia	1.5 million or one fourth of the entire pop ⁿ lost
6	1877-1878	China	9.5 million persons perished
7	1891-1892	Russia	27 million peoples affected
8	1974 & 1987	East Africa	South Sudan and North-East Ethiopia



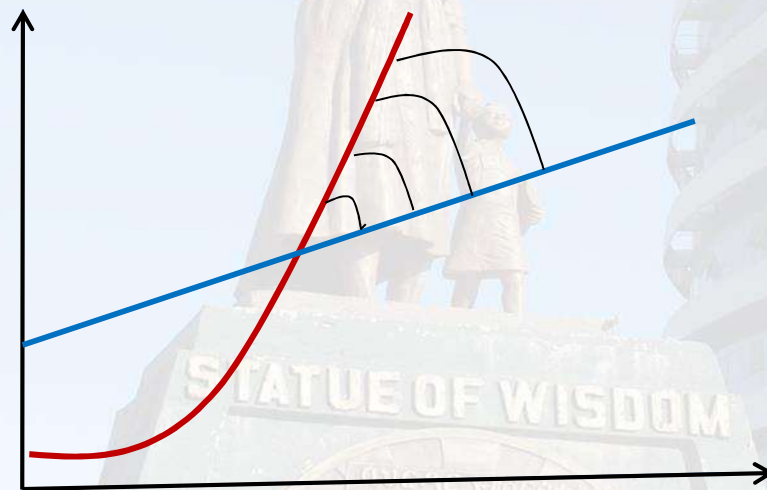
World population growth



Year (AD)

- In relation to population growth and food production, there are two contrasting views:

1. **Alarmist or Neo-Malthusian view** – population increases with geometric means/rate, while crop production increases with arithmetic means/rate that results in subsistence living which leads to misery and degradation

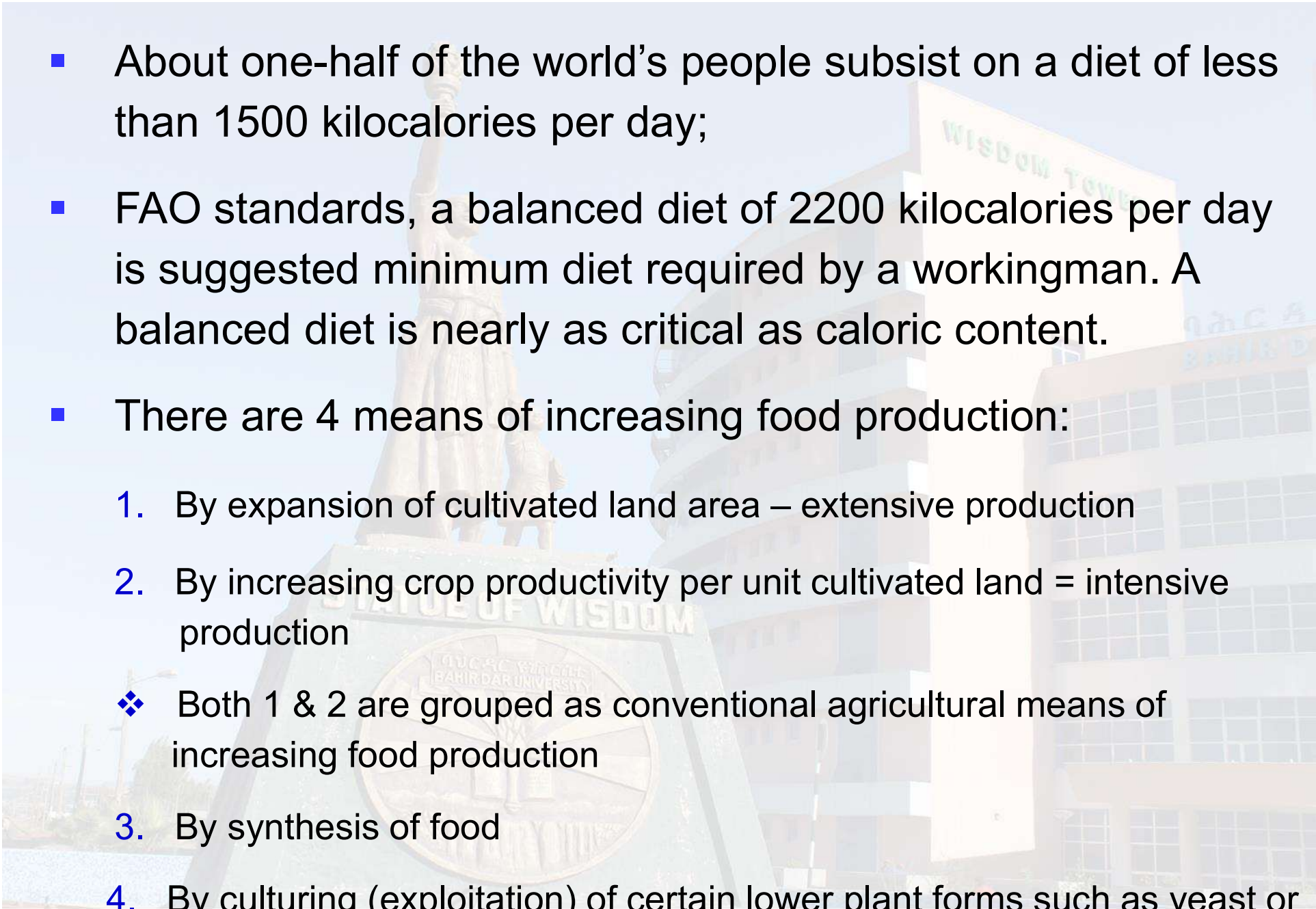


2. **Technocrats view** – food production and supply for the ever increasing population would be solved by technology, and technology allows acceptable level of living for indefinite future

- It is also possible to conclude that **population growth** is a problem, but also it is the cause of advancement and civilization
- Currently there is no deficiency of food production globally
- But, the main problem of food production in the world is the distribution
- There is plenty of food production in developed countries, but there is deficiency of food production in developing countries

Global share of developed & developing countries in population and food production

Countries	Global Share	
	Population	Food Production
Developed countries	30%	55%
Developing countries	70%	45%

- 
- About one-half of the world's people subsist on a diet of less than 1500 kilocalories per day;
 - FAO standards, a balanced diet of 2200 kilocalories per day is suggested minimum diet required by a workingman. A balanced diet is nearly as critical as caloric content.
 - There are 4 means of increasing food production:
 1. By expansion of cultivated land area – extensive production
 2. By increasing crop productivity per unit cultivated land = intensive production
 - ❖ Both 1 & 2 are grouped as conventional agricultural means of increasing food production
 3. By synthesis of food
 4. By culturing (exploitation) of certain lower plant forms such as yeast or Chlorella, which are of highly efficient in net dry matter production



Classification of crop plants



- Crop plants are classified in different ways depending upon the criteria used
- The commonest methods of crop plants classification are:
 1. Agronomic classification – based on the crop use
 2. Growth habit classification – based on the crop life cycle
 3. Climate requirement classification – based on the crop climatic need
 4. Planting habit/cultivation mode classification – based on planting pattern
 5. Botanical classification – based on their morphological similarity of plant parts
 6. Special purpose classification – based on very special purpose of utilization



Agronomic classification



- A. Cereal or grain crops** –include wheat, oats, barley, rye, rice, maize, grain sorghum, millets, teff, and etc. Grain is a collective term applied to cereals.
- B. Pulses** - include faba bean, field pea, chickpea, haricot bean, cow pea, grass pea, lentils, mung beans, soybeans, etc.
- C. Forage crops** include grasses, legumes, crucifers, and other crops cultivated and used for hay, pasture, fodder, silage, or soilage
- D. Root crops** include sugar beats, carrots, turnips, sweet potatoes, cassava, and yams
- E. Fiber crops** – which are produced for fiber extraction. They include cotton, flax, & sisal. Broom corn is grown for its brush fiber



Agronomic classification



F. Tuber crops – include the potato and the Jerusalem artichoke. **G.**

Sugar crops – The sugar beet and sugarcane are grown for their sweet juice from which sucrose is extracted and crystallized. **H. Drug crops** – include tobacco, mint, etc

I. Oil crops –include lean seed, soybeans, peanuts, sunflower, safflower, sesame, Nug (Niger seed), castor bean, Jatropha, mustard, rape seed.

J. Rubber crops : eg. Guayule

K. Vegetable crops: include tomato, onion, lettuce, cabbage, carrot, field pumpkins, maize at milk stage, and many of the pulses

L. Fruit crops: mango, banana, Guava, papaya, avocado, grape, apple, pineapple, etc



Special-purpose classification



1. **Cover crops** – are those seeded to provide a cover for the soil.
2. **Green manure crops** – cover crops turned under while still green would be green manure crops.
3. **Catch crops** – are substitute crops planted too late for regular crops or after the regular crop has failed.
4. **Silage crops** – are those preserved in a succulent condition by partial fermentation in a tight receptacle.
5. **Companion crops** – sometimes called **nurse crops** and they are grown with slowly develop crops
6. **Trap crops** – are those planted to attract certain insect or phanerogamous parasites.



Growth habit classification



- Growth habit classification is based on the life cycle.
- A. An annual** – is short lived plant which completes its entire life cycle from seed to seed in a single growing season and then dies
- B. A biennial** – normally utilizes two complete growing seasons to complete its life cycle. Vegetative growth during the first season and then followed by generative growth during the second growing season
- C. A perennial** – has an indefinite life period. It does not die after reproduction, but it continues to grow indefinitely from year to year



Cultivation mode (planting habit) classification



- It is based on system of planting
 1. **Row planted crops** – are those having large size seed or planting material and they are usually planted with especial arrangement both between rows and plants. For instance, maize, groundnut, potato (tuber), and tomato seedlings
 2. **Drill planted crops** – are planted without especial arrangement between plants and they are of small size seeds. Seeds are drilled in rows by especial sowing machinery and they include barley, wheat, sorghum, teff, finger millet, etc
 3. **Broadcast planted crops:** a seed scattering practice by hand without having especial arrangements between rows and plants
- Hence, such classification is not universal, rather it is merely arbitrary



Climate requirement classification



- Crops are classified on the basis of temperature, altitude or latitude

Classification of crops based on climate requirements

Temperature Req.

1. Warm weather/season crops
2. Cool weather/season crops

Altitude Req.

- Lowland crops
- Highland crops

Latitude Req.

- Tropical crops
- Temperate crops

- **Warm season/lowland/tropical** crops are originated in the tropics and they include: maize, sorghum, millet, rice, haricot bean, soybean, groundnut, cotton, sesame, sugarcane, pepper, sweet potato, tomato, avocado etc.
- **cool season/highland/temperate crops** are originated in the temperate regions and they include: barley, wheat, rye, field pea, faba bean, rape seed, flax, potato, cabbage, sugar beet, lettuce, spinach, garlic, etc.



Botanical classification



- based on morphological similarities of plant parts

Kingdom – *Plantae*

Division – *Tracheophyta*

Sub-division - *Pteropsida*

Class – *Angiospermae*

Subclass – *Monocotyledonae* and *Dicotyledonae*

Order – Mostly *Graminales* and *Rosales*

Family – Mostly *Poaceae* (*Graminae*) and *Fabaceae* (*Leguminosae*)

Genus –

Species –

Variety –



- Classify BH-660 and Moti varieties of maize and faba bean, respectively, based on botanical classification from higher level to lower level:

<u>Level</u>	<u>Maize</u>	<u>Faba bean</u>
Kingdom	Plantae	Plantae
Division	Tracheophyta	Tracheophyta
Subdivision	Pteropsida	Pteropsida
Class	Agiospermae	Angiospermae
Subclass	Monocotyledonae	Dicotyledonae
Order	Graminales	Rosales
Family	Poaceae	Fabaceae
Genus	Zea	Vicia
Species	<i>Zea mays</i> L.	<i>Vicia faba</i> L.
Variety	BH-660	Moti



Please classify Quencho and Abay varieties of Teff (*Eragrostis teff*) and field pea (*Pisum sativum*), respectively, based on botanical classification from higher level to lower level:

<u>Level</u>	<u>Teff</u>	<u>Field pea</u>
Kingdom	_____	_____
Division	_____	_____
Subdivision	_____	_____
Class	_____	_____
Subclass	_____	_____
Order	_____	_____
Family	_____	_____
Genus	_____	_____
Species	_____	_____
Variety	_____	_____



Basic science of agronomic practices of crop production

- **cropping system** is used to describe the pattern in which crops are grown in a given area over a period of time and includes the technical and managerial resources that are utilized.
- Cropping systems are classified based on the following criteria
 1. **The distribution of crops in time**, i.e. whether shifting cultivation, continuous cropping, monoculture, or crop rotation is practiced.
 2. **The distribution of the crops in space on the field**, i.e. whether intercropping or sole cropping is practiced.
 3. **The level of management and resources utilized to produce the crop**, i.e. whether production is intensive or extensive.
 4. **The type of crop grown**, i.e. whether orchard, arable cropping, pasturing, forestry, etc. is practiced.



Shifting cultivation

- In this system, the farm is not at a permanent location.
- Instead, a piece of land is cleared, farmed for a few years and then abandoned in preference for a new site.
- While the new site is being farmed, natural vegetation is allowed to grow on the old site.
- Eventually, after several years of bush fallows, the farmer returns to the original location.
- **Factors necessitating shifting cultivation**
 - ☐ Declining of soil fertility and increasing population.
 - ☐ Unusually high incidence of diseases and pests.
 - ☐ Social or religious customs may dictate the abandonment of site before its fertility level has become marginal.



Continuous cropping

- In contrast to shifting cultivation, continuous cropping implies the cultivation of the same piece of land year after year.
- Fallowing may occur, but it never occurs more than a season or two.
- It should be supported by technology to solve nutrient and insect pest problem like fertilizer application, chemical application, crop selection etc
- **Agricultural practices for maintaining soil fertility under continuous cropping**
- Application of fertilizers and other soil amendments in order to boost fertility.
- Judicious selection of the crops and crop combinations to be grown.
- Introducing short term fallow periods in to the cropping cycle.



Mono-cropping/Sole cropping

- Mono-cropping /Sole cropping is a practice of growing one crop variety alone in the field in normal density. It can be monoculture or crop rotation.
- a. **Monoculture:** this is the practice of repetitive cultivating the same type of crop on the same piece of land year after year. For example sugarcane farming.
- b. **Crop rotation**
 - It is a system of growing different crop species one at a time in definite sequence (in recurrent succession) on the same piece land, season after season or year after year.
 - Importance of crop rotation
 - ✓ Improve soil fertility
 - ✓ Reduce insects and disease infestation
 - ✓ Reduce infection of some weed species.



Intercropping

- Intercropping is the growing of two or more crops simultaneously on the same piece of land per season or year. It is not necessary to harvest the two crops together.
- It may be planted together or at different times .
- **Types of intercropping**
 1. **Row intercropping:**-This follows specific arrangement (row planting) E.g. Maize and bean.
 2. **Mixed intercropping:** - two or more crops are mixed up and broadcasted over the field. Example, Wheat and barley, horse bean and pea etc.
 3. **Relay intercropping:** - is growing of two or more crops simultaneously during the part of the life cycle of each. Succeeding crop is planted before harvesting of preceding crop just after the first crop reached to its reproductive stage, but before harvest.



Intercropping

4. **Strip intercropping** is growing of two or more crops simultaneously in different strips wide enough to permit independent cultivation, but narrow enough for crops interact agronomically.

5. **Multistoried/multilevel/multi-tier intercropping** is the practice of growing different crops of varying height, rooting pattern and duration on the piece of land.

- **Disadvantage of intercropping**
- Since many crops exist together on the field, it is not possible to tailor production practices to the needs of any particular crop.
- Control of pests and diseases is particularly difficult because pesticides which have been developed to control a disease on one particular component crop may have deleterious effect on other crops in the combination.
- It is difficult to mechanize operations such as planting, weeding and harvesting.



Advantage Intercropping

- The spread of diseases and pests is less rapid than in sole cropping.
- There is yield stability.
- Intercropping allows for a more uniform distribution of labour throughout the year.
- Certain crops may exert specific beneficial effect on others.
- The component crop may complement each other in their use of space.
- The crops may complement one another in their use of field time.
- There is yield advantage in growing crops together than growing each one separately.

Alley cropping

- Alley cropping is a cropping system in which crops are grown in the trees or shrubs mainly to hasten soil fertility and control soil erosion.
- In other hands, alley cropping is a form of simultaneous fallow in which quickly growing trees, shrubs or grasses are planted in rows and are regularly cut back;
- the pruning are used as mulch or worked in to the soil in the alleys between the rows.
- **Advantages of alley cropping**
 - ❖ It has a positive influence on the properties of the soil.
 - ❖ It helps to maintain the organic matter status of arable soils
 - ❖ It may serves as a source of food and energy for the soil microbial.
 - ❖ Many alley crops have additional use as sources of food, feed and fuel

Crop Production/Distribution Function

- Crop production and distribution is governed by 4 major factors:
 1. Environmental factors
 2. Socio-economic factors
 3. Crop factors
 4. Management factors (cultural practices)
- Mathematically it can be represented that crop prod. = $f(E, S, G, C)$
where: E=Environmental factors, S= socio-economic factors, G=genetic factors/crop factors, C=cultural practices
- E, S and C are external factors (**extrinsic** factors), while crop factors – G are internal factors (**intrinsic** factors)
- Simply it is also possible to conclude that crop production is governed by two factors: **extrinsic** and **intrinsic** factors, crop production = $f(E, I)$

where: E= extrinsic factors, I= intrinsic factors

1 Some socio-economic factors affecting crop production and distribution are:

- consumption preference
- living standard
- transportation
- income level/source of income
- level of development
- marketing mainly market access and price
- labour availability
- technical/economical feasibility and affordability, etc.

2 Crop factors affecting crop production:

1. Hereditary/genetic factors
2. Seed quality

2.1. Seed quality factors:

- viability/germination
- purity
- healthiness
- size
- uniformity (size and shape)
- freeness from seed-born diseases
- freeness from weeds, especially from noxious weeds
- freeness from mixture of other crop seeds

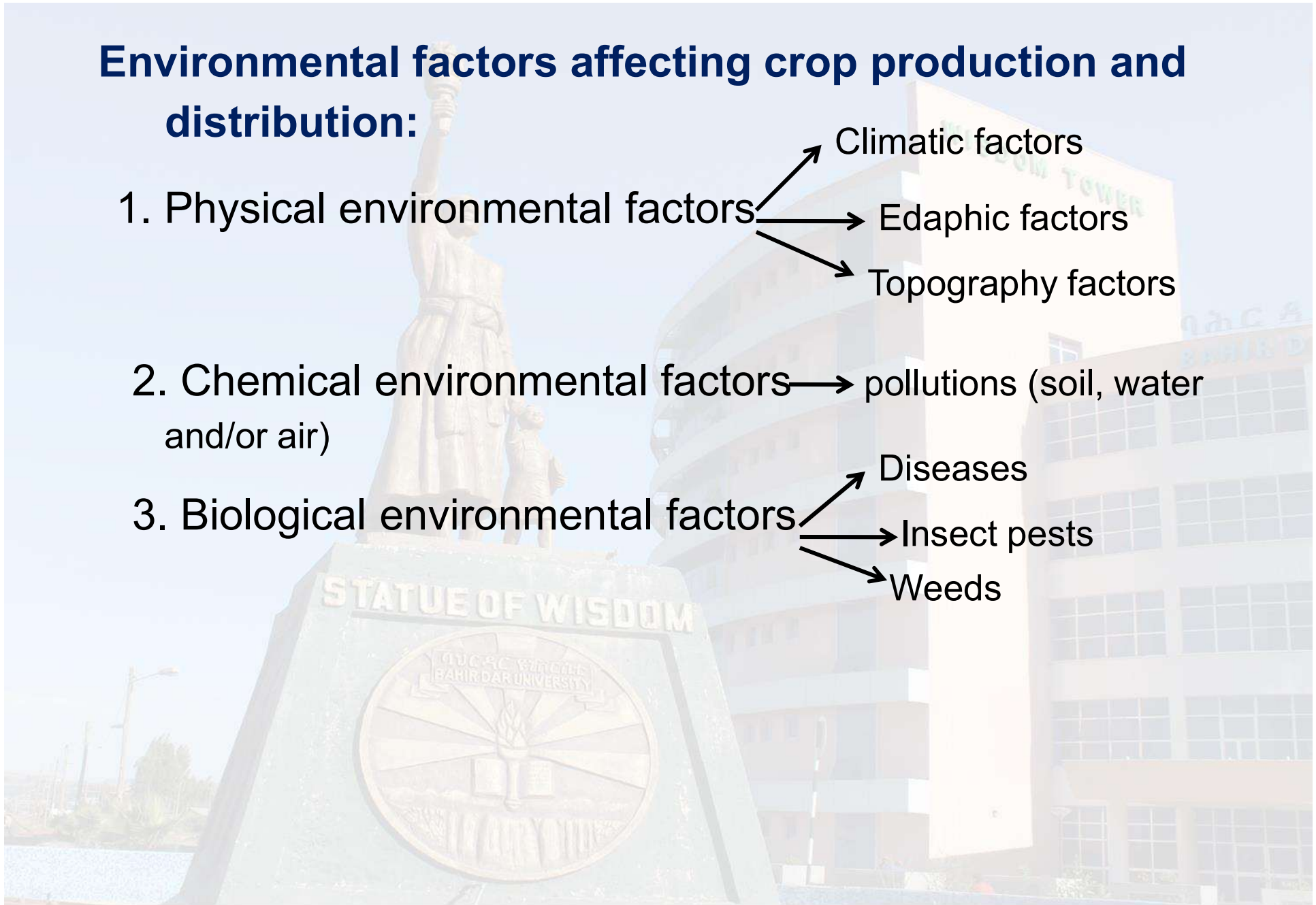


2.2 . Important hereditary/genetic factors considered for selection of crops/varieties:

- yield potential or productive capacity
 - market quality of their produces
 - adaptation to environments
 - resistance/tolerance to abiotic and biotic stresses
 - growth habit, i.e. architecture of a crop
 - effects on soil health, or on succeeding crops
 - harvesting quality
 - animal feeding quality (palatability, nutritive value, texture, etc.)
- Any crop improvement and breeding programs should consider all these factors

Environmental factors affecting crop production and distribution:

1. Physical environmental factors
 - Climatic factors
 - Edaphic factors
 - Topography factors
2. Chemical environmental factors → pollutions (soil, water and/or air)
3. Biological environmental factors
 - Diseases
 - Insect pests
 - Weeds



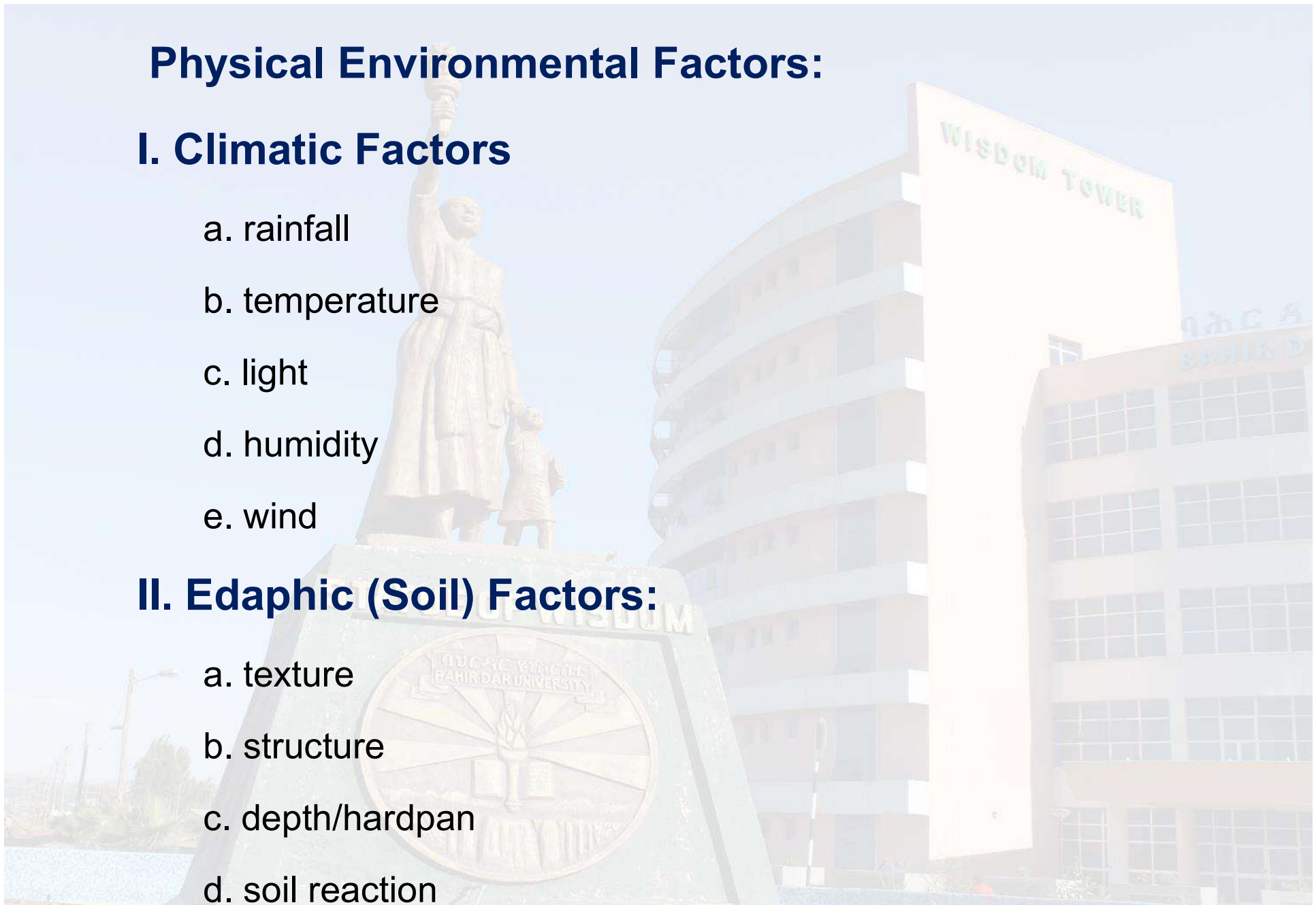
Physical Environmental Factors:

I. Climatic Factors

- a. rainfall
- b. temperature
- c. light
- d. humidity
- e. wind

II. Edaphic (Soil) Factors:

- a. texture
- b. structure
- c. depth/hardpan
- d. soil reaction



e. fertility

f. cation exchange capacity (CEC)

III. Topography:

a. relief

b. slope

Chemical Environmental Factors

- Chemical environmental factors are mainly associated with chemical pollutions including acid rain, pesticides, fertilizers leaching into water bodies, etc.

Biological Environmental Factors

- Biological environmental factors include mainly weeds, diseases and insect pests
- **Environmental stresses** are differentiated into two:
 - a. **Physical stresses** including drought, water logging, chilly, hot temperature, etc. ,

- 
- b. **Biological stresses** including weeds, diseases and insect pests infestations

Management Factors (Cultural Practices):

- a. site selection
- b. land preparation – land clearing, primary tillage, secondary tillage, inter-tillage
- c. soil and water conservation
- d. drainage system/scheme
- e. cropping system – Rain fed vs. irrigation; mono cropping vs. rotation; sole cropping vs. mixed cropping; single cropping vs. multiple cropping
- f. soil fertilization
- g. seeding

Basic practices of crop production

- **Agricultural Practices:** various tasks performed by farmers for crop production are termed as agricultural practices.
 - i. Preparation of the soil
 - ii. Sowing
 - iii. Fertilizer application
 - iv. Irrigation
 - v. Protection measures
 - vi. Harvesting
 - vii. Storage

i. Preparation of the soil

- The preparation of soil is the first step before growing a crop.
- One of the most important tasks in agriculture is to turn the soil and loosen it.
- Activities of soil preparation:
 - **Ploughing:** the process of loosening and turning of the soil is called tilling or ploughing. This is done by using a plough. Ploughs are made of wood or iron.
 - **Watering:** the process of giving water to the field to make the dry soil wet before ploughing.
 - **Levelling:** flattening of the soil after ploughing and watering. The field is levelled for sowing as well as for irrigation processes.

ii. Sowing

- Sowing is the most important part of crop production. before sowing, good quality seeds are selected.
- Sowing is scattering of seeds on the earth or field after soil is prepared for cultivation.
- Cereal grains, grain legumes and forage crops and oil crops are generally direct- seeded, that is the seeds are placed where they will grow and develop to maturity.
- Transplanting, where seedlings are raised in the nurseries and then transplanted to permanent location in the field.
Eg. Most vegetables & trees

Cot'd

- After selecting the appropriate cultivars and preparing the seedbed, major decisions should be made towards crop establishment. These are:-

1. **Depth of seed placement**

2. **Determining optimum sowing time**

3. **Deciding recommended seedling rate:** The seed needed to plant an area is estimated as a weight, not a count of seed.

- Weight of seed /unit area (seed rate) = $(A / N \times \% G \times \% P)$

Where: A= Plants per hectare or per unit area desired

N= Number of seeds per unit weight

P= Purity percentage

G= Germination percentage

Sowing Methods

1. Broadcasting:- the seeds are spread uniformly over well prepared land by plowing or planking.

- It may be done by hand or mechanically spreader. It is suitable for close planted that do not require specific crop or plant geometry.
- **Limitation of broadcasting:**
 - ✓ When seeds fall at different depths it results in the uneven stand.
 - ✓ Difficult to manage, like weeding, hoeing, Earthing up etc
 - ✓ Wastage of fertilizer and seeds

Sowing Methods

2. Drilling: - is the practice of dropping the seeds in the rows or lines. Furrows at specified distance, covered with soil.

- **Advantage of drilling**

- ✓ Seeds and fertilizer both can be drilled together with the use of seed-cum fertilizer drill
- ✓ Maintain a uniform plant population
- ✓ Reduce seed rate
- ✓ Reduce plant competition

- **Limitation of drilling:** - Require more time, more energy and more cost

Sowing Methods

3. Dibbling or planting:- It consists of putting or placing individual seed or seed materials in a hole or pit, made at pre-determined spacing and depth by manual or with the use of mechanical dibbler or planter.

- **Advantage of dibbling**

- ✓ It is important for bigger sized seeds
- ✓ Requires less seeds and gives rapid uniform germination with good seedling vigor.

- **Limitation:** It is more laborious, time consuming and expensive compared to both broadcasting and drilling.

iii. Fertilizer application

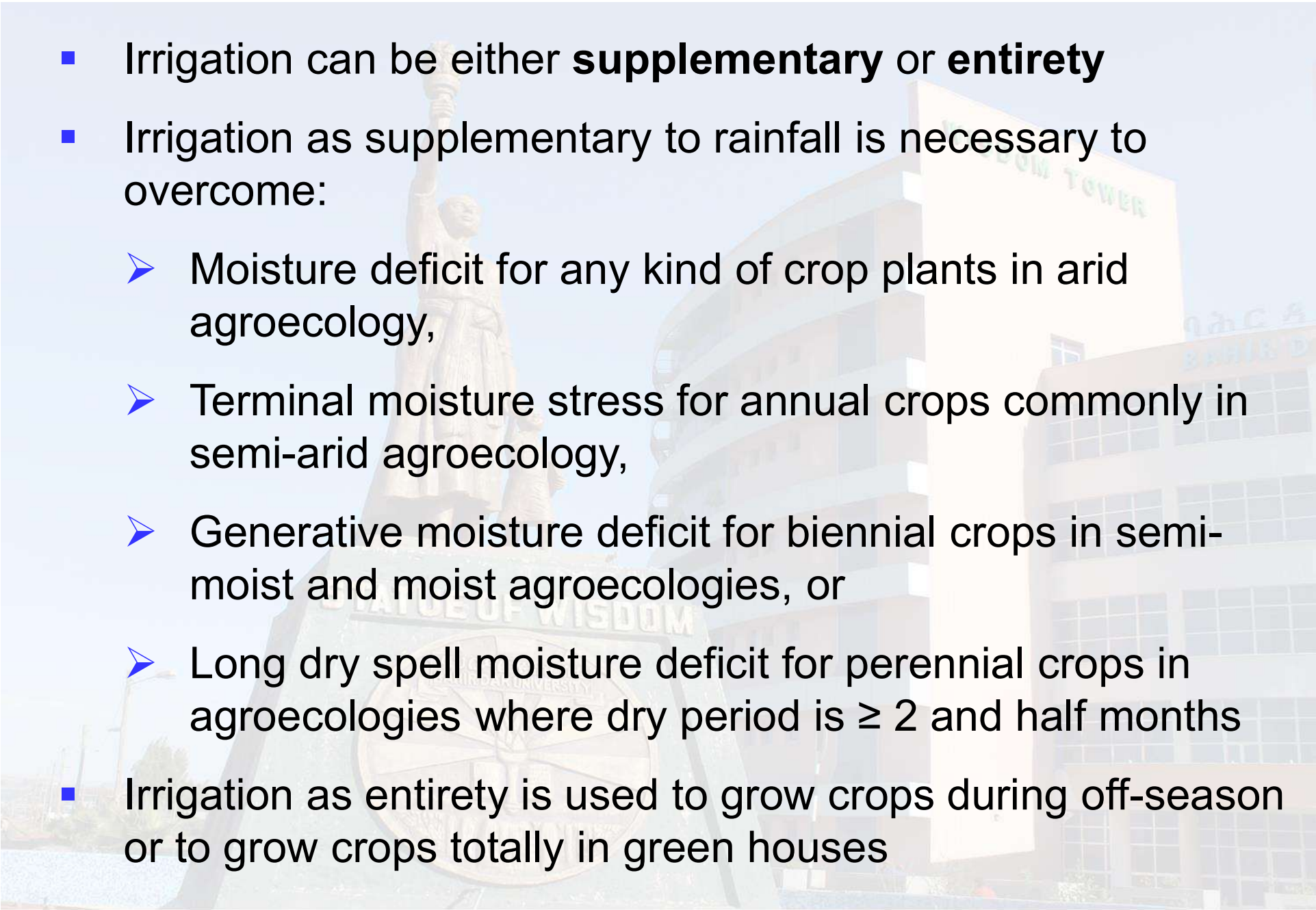
- The substances that are added to the soil in the form of nutrients for the healthy growth of plants are called fertilizers.
- Manure is organic substance obtained from the decomposition of plants or animals waste.
- Fertilizers are chemicals substances which are rich in a particular nutrients.
- Examples of fertilizers are urea, ammonium sulphate, superphosphate, potash, NPK(nitrogen, Phosphorus & Potassium)

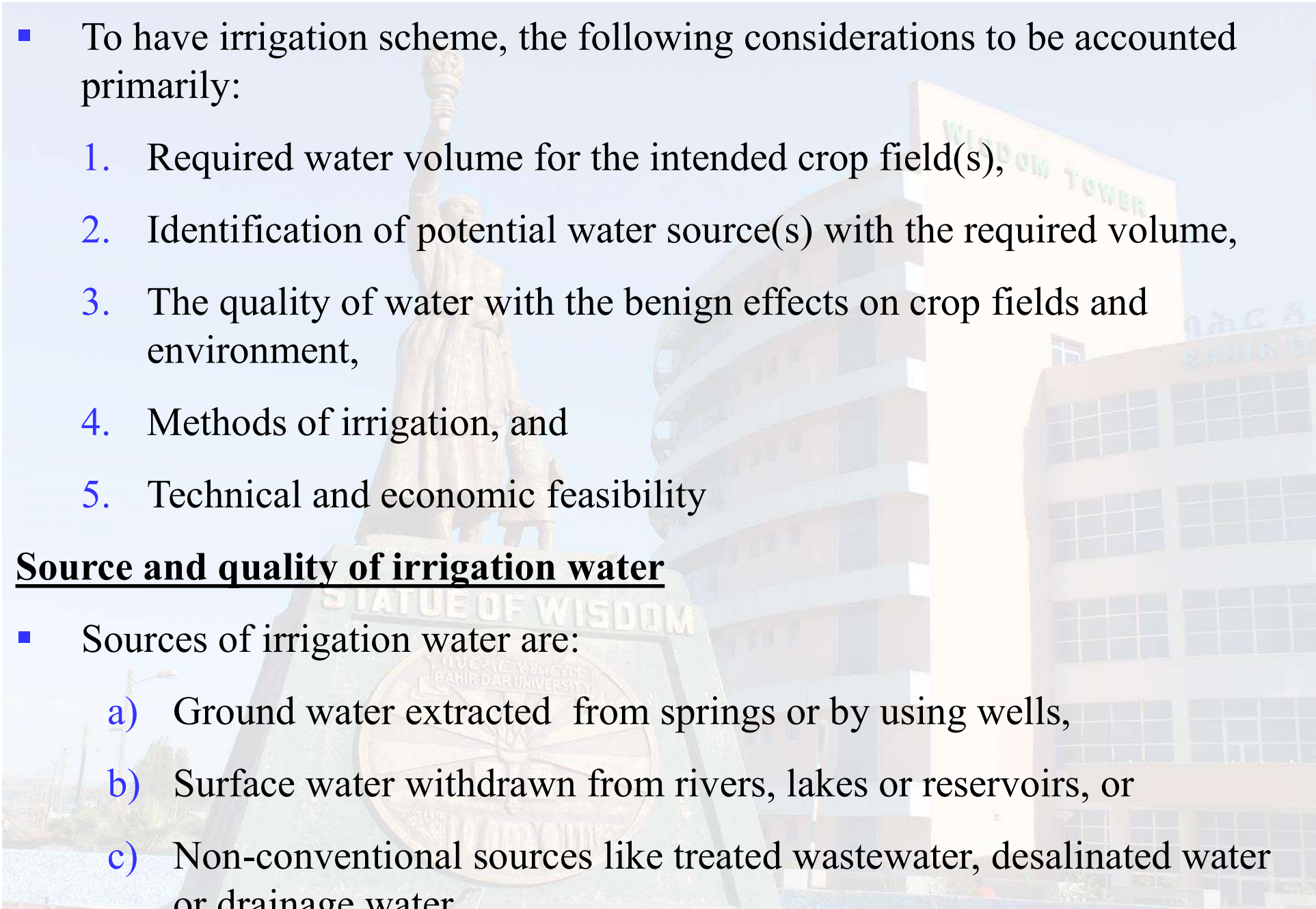
Methods of Fertilizer Application

- **Broadcasting:** -application of fertilizer uniformly on the soil surface before or at the time of sowing and/or in the standing crops. However, due to non uniform distribution and mixing of fertilizer material in the soil, fertilizer use efficiency is low.
- **Band placement:** - application of fertilizer in narrow beneath and/ or by the side of crop rows is known as band placement. In band placement, fertilizer use efficiency is more than broadcast.
- **Foliar application:** - it is application of fertilizer through the leaves. Urea, well rotted urine and many of the micronutrients are often used as foliar application.
- **Fertigation:** - application of fertilizer with irrigation like urea fertilizer.

iv. Irrigation

- The supply of water to crops at different intervals is called irrigation.
- The time and frequency of irrigation varies from crop to crop, soil to soil and season to season.
- irrigation also has a few other uses in crop production, which include (a) protecting plants against frost, (b) suppressing weed growing in grain fields and (c) helping in preventing soil consolidation.
- In contrast, agriculture that relies only on direct rainfall is referred to as **rain-fed** or **dryland farming**.

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- The background of the slide features a faded image of a university campus. On the left, there is a large statue of a person holding a torch, standing on a pedestal. To the right of the statue is a multi-story building with a modern architectural style, featuring large windows and a prominent entrance. The text of the slide is overlaid on this background.
- Irrigation can be either **supplementary** or **entirety**
 - Irrigation as supplementary to rainfall is necessary to overcome:
 - Moisture deficit for any kind of crop plants in arid agroecology,
 - Terminal moisture stress for annual crops commonly in semi-arid agroecology,
 - Generative moisture deficit for biennial crops in semi-moist and moist agroecologies, or
 - Long dry spell moisture deficit for perennial crops in agroecologies where dry period is ≥ 2 and half months
 - Irrigation as entirety is used to grow crops during off-season or to grow crops totally in green houses

- 
- To have irrigation scheme, the following considerations to be accounted primarily:
 1. Required water volume for the intended crop field(s),
 2. Identification of potential water source(s) with the required volume,
 3. The quality of water with the benign effects on crop fields and environment,
 4. Methods of irrigation, and
 5. Technical and economic feasibility

Source and quality of irrigation water

- Sources of irrigation water are:
 - a) Ground water extracted from springs or by using wells,
 - b) Surface water withdrawn from rivers, lakes or reservoirs, or
 - c) Non-conventional sources like treated wastewater, desalinated water or drainage water



v. Protections Measures

- In a field many other undesirable plant may grow naturally along with the crop. These undesirable plants are called weeds.
- The removal of weeds is called weeding. Weeding is necessary since weeds compete with the crops plants for water, nutrients, space and light. thus., they affect the growth of the crops.
- Weeds are plants which are grown out of place
- Weeds are the plants other than the crops sown
- Weeds are plants interfere with the objectives of human beings to cultivate specific types of crop.

Major characteristics of ideal weeds

- Ability to flower and produce seeds in a short period of time
- Ability to produce large number of seeds
- Presence of efficient mechanisms for seed dispersal
- Presence of allelo-chemical that inhibit growth of crops
- Possession of specialized seed dormancy mechanisms
- Ability to survive in a wide range of environment
- Ability to resist chemicals
- Have both inter and intra competition behavior

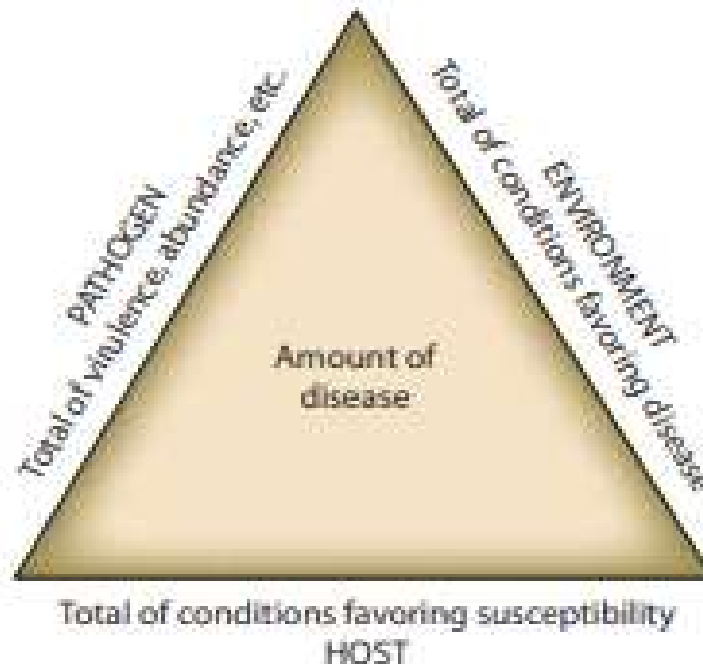
Insect pests

- Insecta (insects) are arthropods that have small size, three body regions (head, thorax and abdomen), three pairs of legs, a pair of antennae and they feed by mandibles..
- ***Beneficial insects***: Pollinators (e.g. Bees, wasps), Parasitoids (e.g. Tricogramma on stalk borer), and Predators (e.g. Lady Bird beetle on aphids).
- ***Harmful insects***: Field pests (e.g. army worm), Storage pests- (weevils, flour moths), and Vectors/disease transmitters (leaf hopper, aphids).
- The major economic insect pests in Ethiopia are African boll worm, cereal stem borer, Red teff worm, sorghum chafer, wello bush cricket, Aphids, Army worm, weevils, flour moths etc.

Plant disease

- **Disease** is harmful deviation from the normal function of physiological process of an organism. Crop diseases cause yield reduction.
- Plant disease may be classified into two categories on the basis of causal agent:
 - Abiotic (non-infectious) disease: caused by environmental factor and thus not infectious. E.g. abnormal levels of growth requirement (high, low or extreme).
 - Biotic (Infectious) disease: - caused by pathogen and can be transmitted. The organisms are fungus, bacteria, nematodes and viruses, parasitic higher plants.
- The interaction between the causal organism (pathogen), host and certain factors within the environment and these factors form what is called the **disease triangle**.

Plant disease



- Disease will occur only when the pathogen interacts with a susceptible host under favorable conditions. The presence of a pathogen alone is not sufficient to cause disease.
- The most serious crop diseases in Ethiopia e.g. Rust, Smut, Leaf blight, Root rot, powdery mildew, Damping off. Etc.

Pest control methods

a. **Preventive methods:** prevention is better the interventions/cure/, it includes using:

- clean seeds,
- well decomposed organic matter,
- clear farm land,
- cleaning irrigation channels,
- burning the straw in the field and
- proper quarantine law.

b. **Cultural methods:-** refers to the use of management practices that make the environment less favorable to pest reproduction, dispersal and/ or survival .

Examples:

- ❖ time of panting,
- ❖ site selection,
- ❖ crop rotation,
- ❖ seeding and planting methods,
- ❖ crop residue and alternative host destruction,
- ❖ tillage practices and appropriate cropping system,
- ❖ fertilizer and water management.

Pest control methods

C. Resistance species/competitive or smoother crop/non attractive cultivars

d. Physical methods:

- ✓ hand-weeding,
- ✓ tillage,
- ✓ mowing,
- ✓ burning weeds and crop residues,
- ✓ hoeing,
- ✓ flooding,
- ✓ mulching,
- ✓ manual collection and destruction of insect pests,
- ✓ mechanical exclusion of pests by means of nets, ditches, fences and traps.

e. Biological methods: use of natural enemies namely predators, parasitoids and pathogen to maintain the density of pest species at lower densities than could occur in their absence.

f. Botanical and animal products/organic substances –use available botanical and animal products are very important for controlling of pests.

Pest control methods

g. Chemical methods: using insecticides, fungicides, herbicides etc.

- Herbicides may be classified based on:
 - **Non-selective herbicides:** kill all plants when applied at adequate rates
 - **Selective herbicides:** kill or stunt weeds in a germinations or growing crop without harming the crop e.g. Gram axon, 2,4-D

h. Integrated Pest Management (IPM): use of two or more than two control methods through integrating.

- It is a pest management strategy that is socio- economic context of farming systems, the associated environment and the population dynamics of the pest species, utilizes all suitable techniques and methods in as compatible manner as possible, and maintains the pest population level below those causing economic injury.

vi. Harvesting:

- Harvesting is the processes of collecting various plant produces, whenever they are ready for utilization or marketing.
- Harvesting should be done at proper stage where there is minimum damage and loss, as rapidly as possible and at minimum cost.
- Harvesting should be done at early morning or late evening hours.
- A temperature of above 27°C during harvesting should be avoided.
- The products that are to be send to distant markets are harvested in the evening and transported in the cool hours of night whereas commodities for local markets are harvested early morning.
- Harvesting should not be done immediately after rain or irrigation.
- Harvesting at optimum stage of maturity ensures maximum quality and yield.
- Care must be taken to avoid mechanical injury to product.

Harvesting:

- **Types of maturity**

- a) **Physiological maturity:** – the stage of development of the product at which maximum dry weight has been attained by the plant and consequently no gains in product yield can occur with increased production inputs. At physiological maturity, moisture content drops up to 20 %.
- b) **Harvest maturity:** - The stage of harvesting a product to obtain peak quality and quantity, as determined by the producer.

- **Methods of harvesting**

- I. **Manual harvesting:** - in most of the developing countries, harvesting is often done manually. In manual harvesting sickle and scythe are most important tools.
- II. **Combine harvesting:-** the mechanical harvesting of a crop product using a combine, that is capable of multiple operations, usually harvesting, shelling and cleaning together.

Postharvest loss

- **postharvest losses** are defined as “losses that occur after harvest till the produce reaches consumers.



vii. Storage

- Store only clean, unbroken kernels should be selected for storage.
- The grain should be dry (low moisture content of products 8-15%) and
- it should be cool when it is put into storage.
- If the crop grains are to kept for longer time, they should be free from moisture, insects, rats and microorganisms.
- Farmers store grains in jute or metallic bags. however, large scale storage of grains is done in silos and granaries to protect from pests like rats and insects.

Group Assignment

1. Cereals crop production (Group 1)
 - Introduction
 - Role of cereals in nutrition
 - Economic role of cereals
 - Cereals production system
 - Conclusion
2. Pulses crop production (Group 2)
 - Introduction
 - Role of pulses in nutrition
 - Economic role of pulses
 - Pulses production system
 - Conclusion
3. Vegetable crops production(Group 3)
 - Introduction
 - Role of vegetables in nutrition
 - Economic role of vegetables
 - vegetables production system
 - Conclusion
4. Fruit crops production (Group 4)
 - Introduction
 - Role of fruits in nutrition
 - Economic role of fruits
 - fruits production system
 - Conclusion
5. Root & tuber crops production (Group 5)
 - Introduction
 - Role of Root & tuber crops in nutrition
 - Economic role of Root & tuber crops
 - Root & tuber crops production system
 - Conclusion



Thank you for your attention!